Cross-Sell Analysis

Transforming 1.73M+ transactional records into actionable insights for multi-product growth strategy

Made with GAMMA

PROJECT OVERVIEW

Scaling Efficiency & Cross-Sell Health

This project analyzes the scaling efficiency and cross-sell health of a multi-product operation by processing a 1.73M+ record database across 6 relational tables. The primary goal was to transform raw transactional data into 32,313 verified orders to identify high-revenue product bundles.

171.83%

YoY Revenue Growth

Massive revenue surge year-over-year

$110K

Revenue Leakage

Total refund impact to address

32,313

Verified Orders

Cleaned dataset for analysis

This analysis isolates the most profitable cross-sell pairings to guide global marketing and quality control strategies.

Made with GAMMA

DATA FOUNDATION

Dataset Summary

Scale & Volume

Processed **1.73M+ raw records** across **36 distinct data fields** spanning 6 tables, covering July 2012 to January 2015. SQL filtering aggregated this into **32,313 verified orders** for the final reporting model.

Financial Metrics

Captured **$1.94M in Gross Revenue** and tracked **$110.04K in total refund impact** to understand profitability dynamics.

Behavioral Insights

Analyzed **8,000 cross-sell units**, establishing a benchmark **23.89% attachment rate** while monitoring **1,731 refunded units** for quality control.

Database schema: fuzzy_factory with 6 relational tables

Made with GAMMA

DATA PROCESSING

ETL Pipeline using SQL

A multi-stage ETL process was developed to handle high-volume transactional data and ensure dashboard accuracy.

Extraction

Queried **1.73M+ raw records** from source database, mapping all 36 data fields correctly for analysis.

Transformation

Performed complex SQL queries to aggregate transactions into meaningful business metrics for performance tracking.

Validation

Rigorous filtering to arrive at **32,313 verified orders**, maintaining data integrity for all KPIs and visualizations.

Load

Structured final datasets into optimized views for seamless Power BI integration.

```
create view prd_performance_fact as
select o.order_id, o.created_at, o.primary_product_id, o.price_usd,
case when i.order_id is not null then 1
else 0 end as is_cross_sell,
case when r.order_id is not null then 1
else 0 end as is_refunded
from orders o
left join order_items i on o.order_id = i.order_id
and i.is_primary_item = 0
left join refunds r on o.order_id = r.order_id;
```

Final SQL transformation bridging 1.73M raw rows into ~32K analytical dataset

Made with GAMMA

KEY INSIGHTS

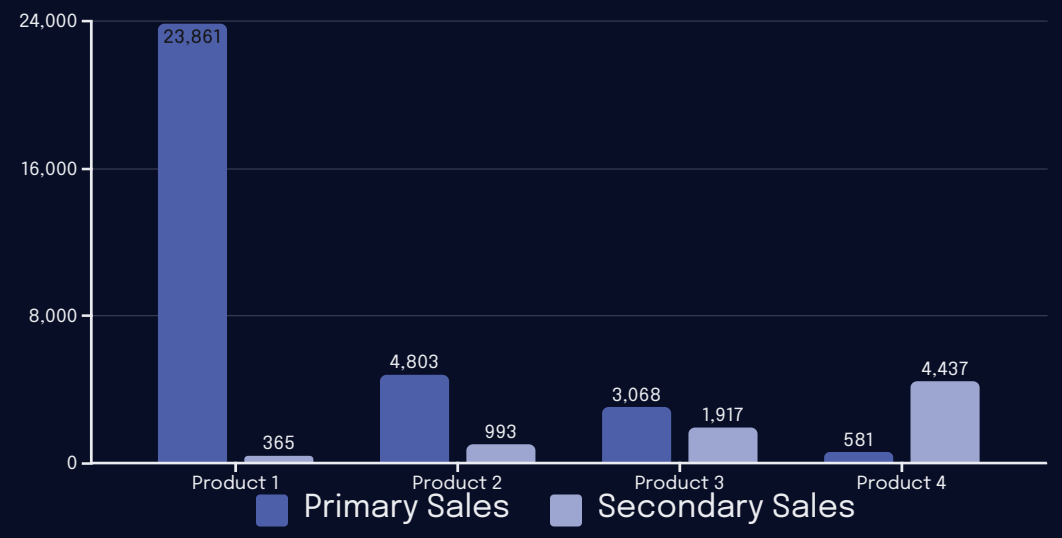
Product Distribution Analysis

Primary vs. Secondary Sales

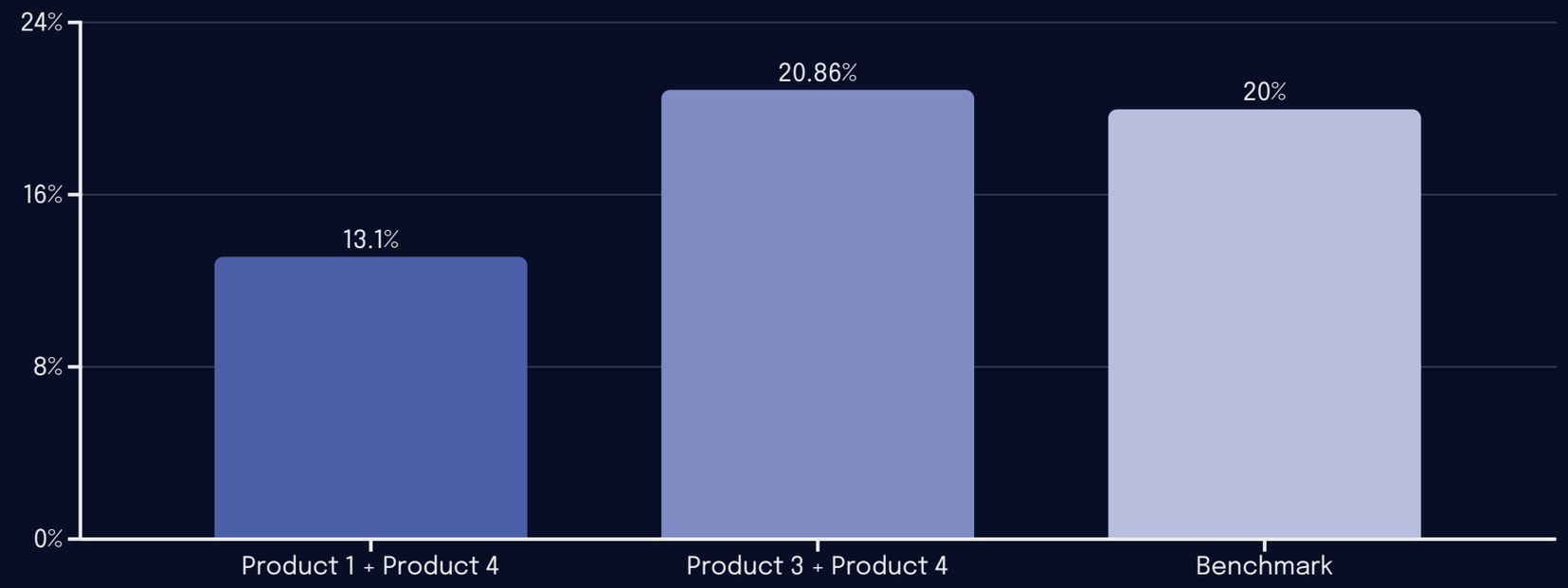
Analysis of market demand reveals how primary products drive business versus secondary attachments.

Product 1 dominates with **23,861 primary units** and 365 secondary units, confirming its role as the volume leader.

Product 4 shows inverse behavior: only **581 primary units** but an impressive **4,437 secondary units**, making it the ultimate cross-sell product.



Made with GAMMA



Cross-Sell Performance Deep Dive

Best Drive Product

Product 1 drives the most Product 4 sales with **3,126 drive sales**, leveraging its massive volume advantage.

Attachment Rate Gap

Product 1 + Product 4 achieves only **13.10% attachment**, below the 20% benchmark, indicating untapped potential.

Highest Performer

Product 3 is the most effective drive with a peak **20.86% attachment rate**, making it ideal for bundled deals.

Key Finding: While Product 1 has the highest volume, Product 3 demonstrates superior cross-sell efficiency, suggesting different marketing strategies for each.

Made with GAMMA

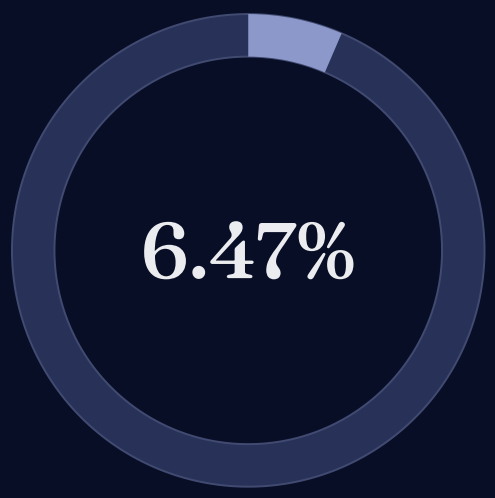
RISK ASSESSMENT

Refund Rate & Quality Analysis

Bundle Risk Evaluation

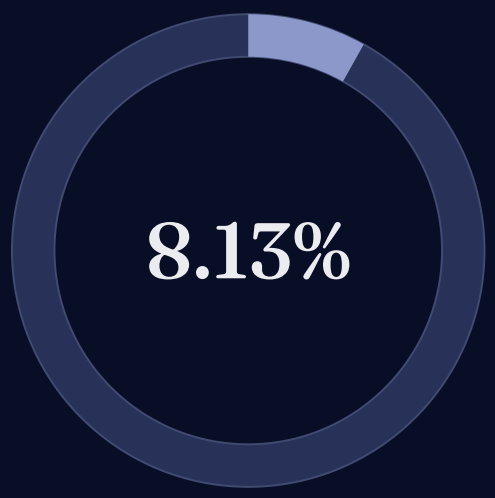
Since Product 3 is the primary growth driver, refund analysis is critical. The **Product 3 + Product 4** combo shows an **8.13% refund rate**, higher than Product 3 alone (6.47%).

This elevated refund rate suggests the bundle may introduce quality or customer expectation risks that require investigation before scaling marketing spend.



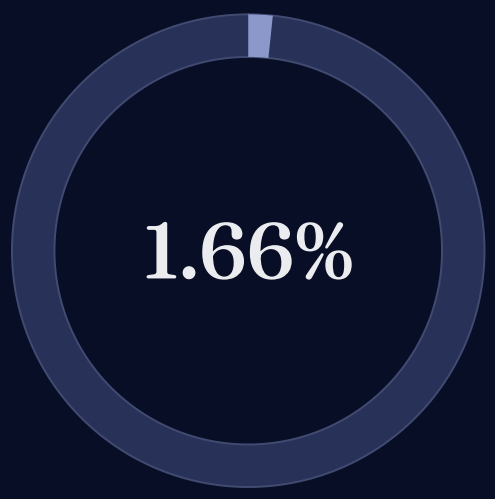
Product 3 Alone

Baseline refund rate



Product 3 + 4 Bundle

Elevated bundle risk



Risk Increase

Additional bundle exposure

Made with GAMMA

Revenue Maximization Strategy

Analysis of Average Order Value (AOV) across combinations identifies the most profitable bundles for revenue optimization.

Product 2 + 4

$89.98 Combo AOV

Highest per-order value

Product 1 + 4

$79.98 Combo AOV

High volume bundle

Product 3 + 4

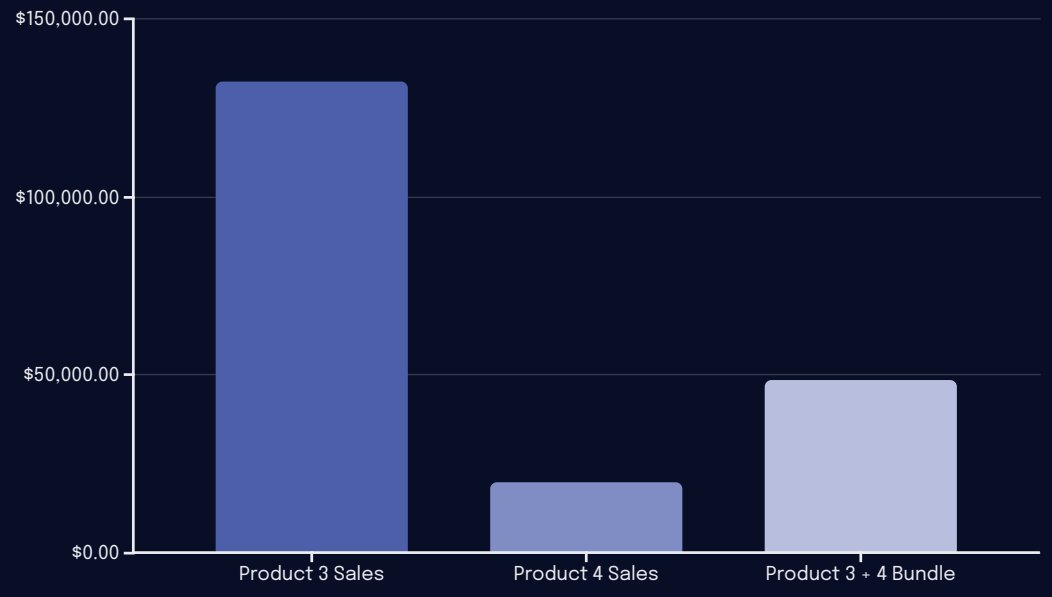
$75.98 Combo AOV

Best attachment rate

Bundle Revenue Impact

The **Product 3 + Product 4** bundle generated **$48,627** in total revenue, proving that bundling increases revenue despite the $110.04K total refund impact across the company.

Individual Product 3 sales: **$132,230**
Individual Product 4 sales: **$19,776**
Combined bundle revenue: **$48,627**



Made with GAMMA

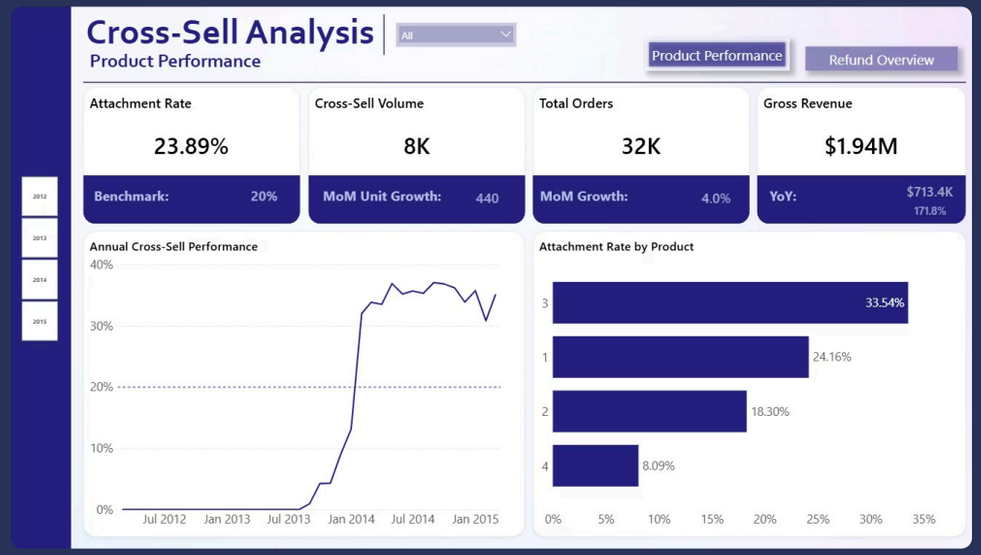
INTERACTIVE DASHBOARD

Power BI Dashboard Overview

Built comprehensive interactive dashboard showing product performance and refund analysis with real-time KPI tracking.

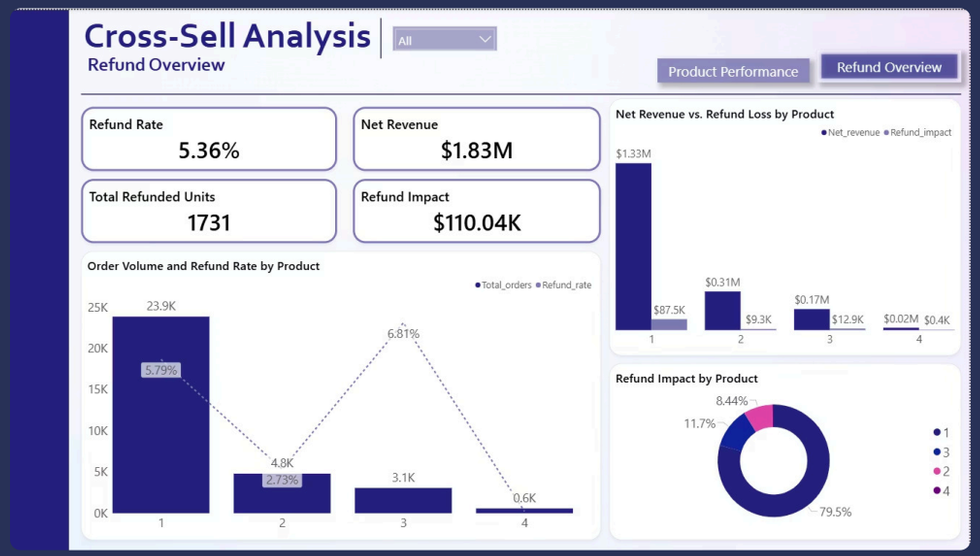
Product Performance

This dashboard tab provides a high-level overview of total sales, gross revenue (**$1.94M**), and order volume. It helps track YoY growth (**171.8%**) and identify general market trends.



Refund Overview

Detailed insights into cross-sell performance, including attachment rates (**23.89%**) and refund rates (**5.36%**) by product bundle. Pinpoint specific product combinations driving success or requiring attention due to high refunds.



Made with GAMMA

STRATEGIC RECOMMENDATIONS

Business Recommendations

01

Scale High-Performance Bundles

Launch targeted marketing for the **Product 3 + Product 4** combo, leveraging its **20.86% attachment rate** as the most effective cross-sell driver.

02

Quality Audit for Product 3

Investigate the **8.13% bundle refund rate** to identify if issues stem from shipping damage or customer expectations before increasing marketing spend.

03

Prioritize Stable Growth

For global expansion with lowest risk, prioritize **Product 2** with its healthy **13.97% attachment rate** and lowest refund risk at **2.73%**.

04

Efficiency Drive for Product 1

Improve cross-sell strategy for Product 1 (currently **13.10%**). Even a 2% increase would drive massive revenue due to its high sales volume of 23,861 units.

Made with GAMMA